

EXP 23 – Family Tree

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Aim

To write a Prolog program that represents family relationships using facts and rules, and to infer relations such as father, mother, grandfather, and siblings.

Algorithm

1. Define male and female individuals.
2. Define parent relationships using parent(X, Y).
3. A person is a **father** if he is male and a parent of the child.
4. A person is a **mother** if she is female and a parent of the child.
5. A person is a **grandfather** if he is male and parent of a parent.
6. Two people are **siblings** if they share at least one parent and are not the same individual.
7. Use forall and format to display all relationships.

Source Code

```
prolog
```

```
male(john).
```

```
male(michael).
```

```
male(david).
```

```
male(james).
```

```
female(susan).
```

```
female(linda).
```

```
female(emma).
```

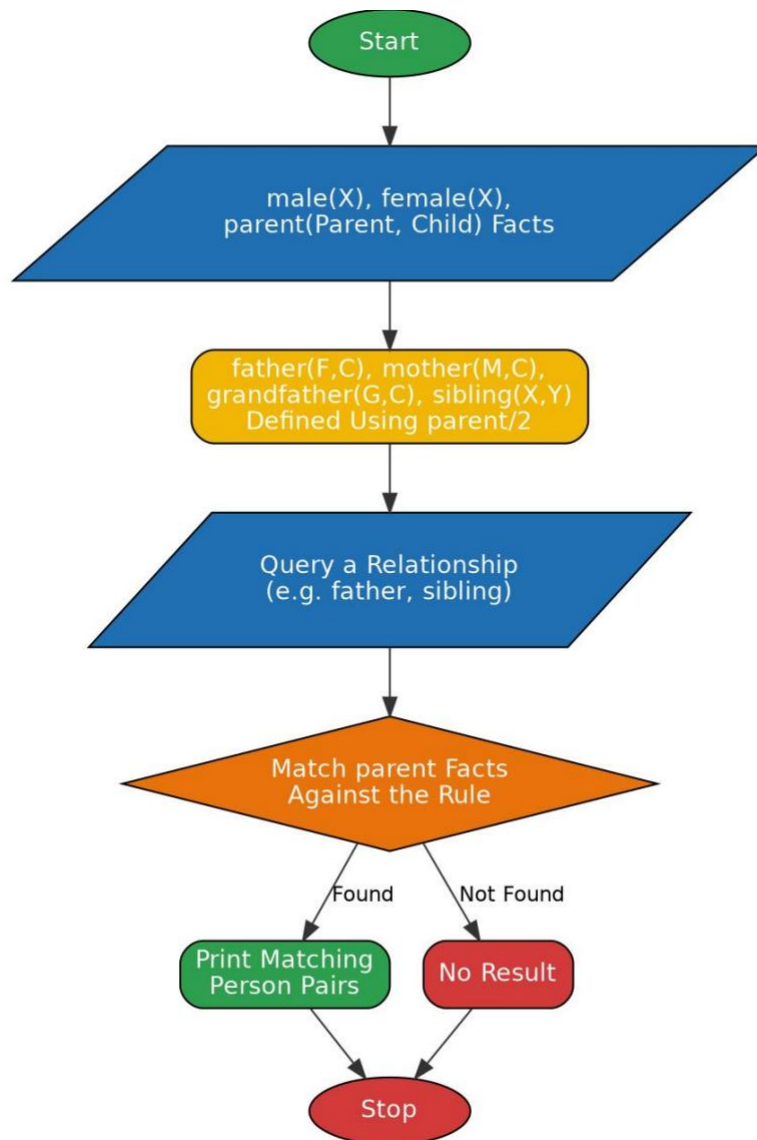
```
female(sophia).
```

parent(john, michael).
parent(john, linda).
parent(susan, michael).
parent(susan, linda).
parent(michael, david).
parent(michael, emma).
parent(linda, james).
parent(linda, sophia).

father(F, C) :- male(F), parent(F, C).
mother(M, C) :- female(M), parent(M, C).
grandfather(G, C) :- male(G), parent(G, P), parent(P, C).
sibling(X, Y) :- parent(P, X), parent(P, Y), X \= Y.

show_fathers :- forall(father(F, C), format("~w is the father of ~w~n", [F, C])).
show_mothers :- forall(mother(M, C), format("~w is the mother of ~w~n", [M, C])).
show_grandfathers :- forall(grandfather(G, C), format("~w is the grandfather of ~w~n", [G, C])).
show_siblings(P) :- setof(X, sibling(P, X), L), forall(member(S, L), format("~w~n", [S])).

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
?- cd('C:/Users/yanar/Documents/CSA1717').
true.
```

```
?- [family].
true.
```

```
?- show_fathers.
john is the father of michael
john is the father of linda
michael is the father of david
michael is the father of emma
true.
```

```
?- show_mothers.
susan is the mother of michael
susan is the mother of linda
linda is the mother of james
linda is the mother of sophia
true.
```

```
?- show_grandfathers.
john is the grandfather of david
john is the grandfather of emma
john is the grandfather of james
john is the grandfather of sophia
true.
```

```
?- show_siblings(michael).
linda
true.
```

```
?- show_siblings(linda).
michael
true.
```

Result

The program successfully identifies and displays all fathers, mothers, grandfathers, and siblings based on the defined family relationships. It correctly infers each relation using logical rules and prints them in a readable format.